Optical Communications

Salwa El-Sabban

Assessment

a) Lab + Oral	**20**
b) Year work	**30**
c) Final-term exam	**100**
Total	**150/150**

2 Lect. + 1 Tutorial + 1 Lab/ Week

Course Contents

-Optical fiber fundamentals :

Fiber analysis, single mode and multimode fibers, concept of V number, graded index fibers, attenuation mechanisms in fibers, dispersion in single mode and multimode fibers, dispersion types, dispersion shifted, and dispersion flattened fibers, Erbium-Doped Fiber Amplifier (EDFA).

-WDM components :

Directional couplers, splitters and junctions, optical switches, optical interferometers, optical filters, ring resonators, multiplexer, optical modulators.

-System:

Coupling and connector loss, splice loss**, t**ransmitter design circuitry, typical receiver configurations, noise in optical receivers, power budget, rise time budget.

Optical networks: optical network topologies, WDM network (DWDM, CWDM), Passive optical networks (GPON and EPON), Free Space Optics (FSO) communication systems, Visible Light Communication (VLC).

Text Books

1- Gerd Keiser, "Fiber optic communications", Springer, 2021
2- Shiva Kumar and M. Jamal Deen, "Fiber optic communications :fundamentals and applications" , Wiley, 2014
3-Bob Chomycz, "Planning fiber optic networks", McGraw Hill, 2009
4- R. Ramaswami, K. N. Sivarajan, G. H. Sasaki, "Optical networks", Morgan Kaufmann :Elsevier, 2010
5- Gerd Keiser, "Optical fiber communications", McGrawv Hill, 2012
6- Gerd Keiser, "FTTX concepts and applications", A John Wiley & Sons Inc., 2006

WHY OPTICAL FIBER?

- **High information capacity (GB/s)**
- **Very long distances**
- **Strong, flexible, and reliable**
- **Light weight cables**
- **Secure**
- **Immune to EM interference (EMI)**
- **Chemically stable (no corrosion)**
- **Low Thermal Expansion**

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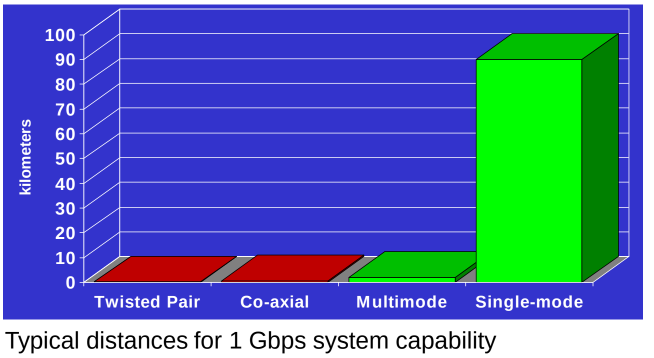
FIBER ➔ LONGER DISTANCES*



* Typical distances for 1 Gbps system capability

Source: Corning Incorporated

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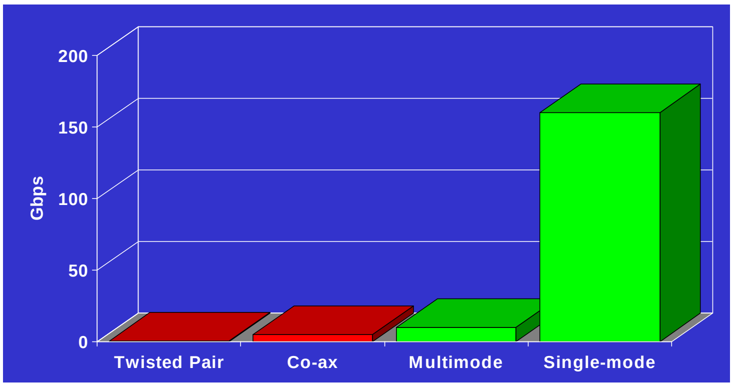
FIBER ➔ MORE CAPACITY*



* Typical system capability for 100 m link

Source: Corning Incorporated / FTTH Council

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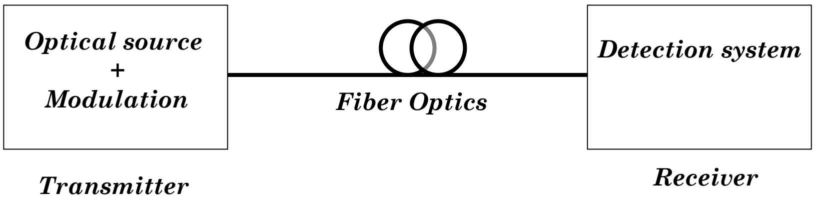
7

Typical point-to-point optical communication system



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8

System Performance

- Repeater-less fiber length L
- Capacity = Bandwidth B

  Figure of merit = LxB
- Parameters :
  - Fiber Attenuation & Dispersion
  - LD modulation Bandwidth and spectrum
  - PhD sensitivity and noise
  - Detection techniques
  - …etc.

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9

Telecom Windows

Optical fiber communications typically operate in a wavelength region corresponding to one of the following "telecom windows":

•The first window at 800–900 nm was originally used. GaAs/AlGaAs-based laser diodes and light-emitting diodes (LEDs) served as transmitters, and silicon photodiodes were suitable for the receivers.

However, the fiber losses are relatively high in this region, and fiber amplifiers are not well developed for this spectral region. Therefore, the first telecom window is suitable only for short-distance transmission.

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10

Telecom Windows

•**The second telecom window utilizes wavelengths around 1.3 µm, where the loss of silica fibers is much lower and the fibers chromatic dispersion is very weak, so that dispersive broadening is minimized.**

This window was originally used for long-haul transmission. However, fiber amplifiers for 1.3 µm (based on praseodymium-doped glass) are not as good as their 1.5-µm counterparts based on erbium.

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11

Telecom Windows

•**The third telecom window, which is now very widely used, utilizes wavelengths around 1.5 µm. The losses of silica fibers are lowest in this region, and erbium-doped fiber amplifiers are available which offer very high performance. Fiber dispersion can be tailored with great flexibility (→ dispersion-shifted fibers).**

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12

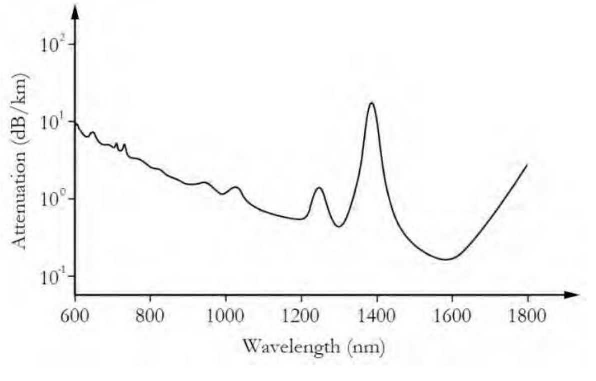
Telecom Windows

The second and third telecom windows are further subdivided into the following wavelength bands:

Band	Description	Wavelength range
O band	original	1260–1360 nm
E band	extended	1360–1460 nm
S band	short wavelengths	1460–1530 nm
C band	conventional ("erbium window")	1530–1565 nm
L band	long wavelengths	1565–1625 nm
U band	ultralong wavelengths	1625–1675 nm

The second and third telecom windows were originally separated by a pronounced loss peak around 1.4 μm, but they can effectively be joined with advanced fibers with low OH content which do not exhibit this peak.

Fiber Attenuation



Attenuation of an optical fiber as a function of wavelength

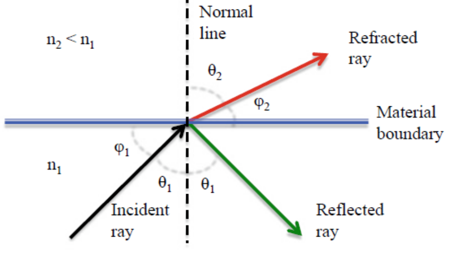
LONG DISTANCE POINT TO POINT SYSTEM

- λ= 850 nm, GaAs Laser source & Si detector
- λ= 1300 nm, InP Laser source & InGaAs detector
- λ= 1550 nm, InP Laser source & InGaAs detector
- λ= 1550 nm, with EDFA
- λ= 1540-1560 nm, with EDFA +DWDM

Basis of Reflection and Refraction

*When a light ray encounters a boundary separating two different dielectric media, part of the ray is reflected back into the first medium and the remainder is bent (or refracted) as it enters the second material.

*The bending or refraction of the light ray at the interface is a result of the difference in the speed of light in two materials that have different refractive indices.



The relationship at the interface is known as *Snell's law* and is given by

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

or

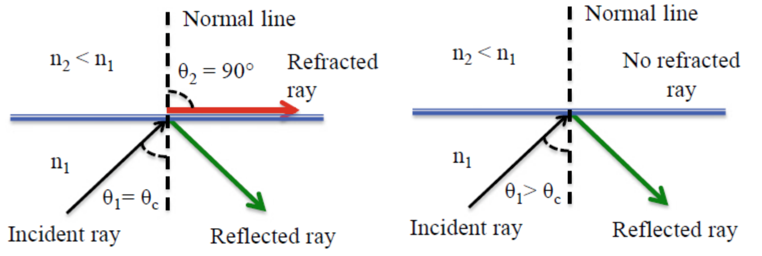
$$n_1 \cos \varphi_1 = n_2 \cos \varphi_2$$

θ_1 is *the angle of incidence*

Basis of Reflection and Refraction

*The incident ray, the normal to the interface, and the reflected ray all lie in the same plane, which is perpendicular to the interface plane between the two materials. This plane is called the *plane of incidence*

*As the angle of incidence θ_1 in an optically denser material ($n_1>n_2$) becomes larger, the refracted angle θ_2 approaches $\pi/2$. Beyond this point no refraction is possible as the incident angle increases and the light rays undergo *total internal reflection*



Θ_c is called *critical angle of incidence*

Basis of Reflection and Refraction

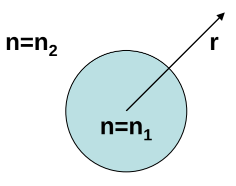
*When the incidence angle θ_1 is greater than the critical angle, the condition for total internal reflection is satisfied; that is, the light is totally reflected back with no light escaping to the n_2 medium.

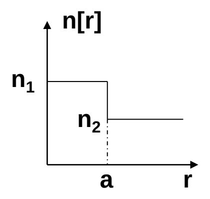
As the critical angle is reached when $\theta_2 = 90°$ so that $\sin\theta_2 = 1$, then from Snell's law

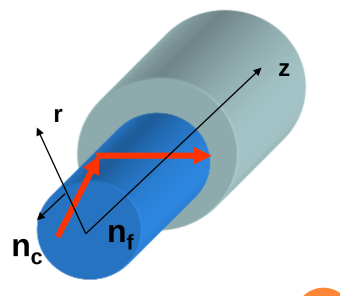
$$\sin\theta_c = \frac{n_2}{n_1}$$

FIBER WAVEGUIDE

- What is a fiber waveguide ?
  - Open Dielectric waveguide using total internal reflection







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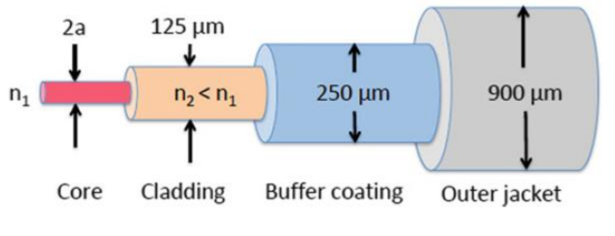
Fiber Waveguide

An optical fiber is a dielectric waveguide that operates at optical frequencies. It is normally cylindrical in form. It confines electromagnetic energy in the form of light to within its surfaces and guides the light in a direction parallel to its axis.



A conventional silica fiber has a circular solid core cross section of refractive index n1 surrounded by a cladding with a refractive index n2 < n1; an elastic plastic buffer encapsulates the fiber

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Fiber Types

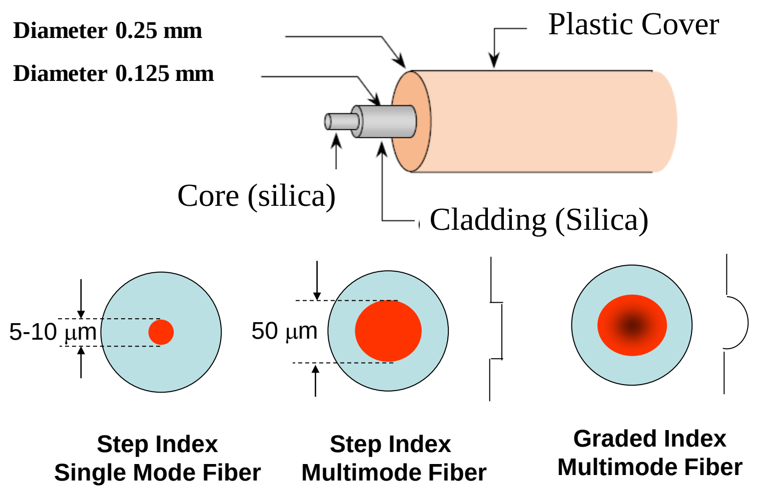
Step-index fiber:
The refractive index of the core is uniform throughout and undergoes an abrupt change (or step) at the cladding boundary.

Graded-index fiber
The core refractive index is made to vary as a function of the radial distance from the center of the fiber.

Single-mode fiber
It sustains only one mode of propagation.

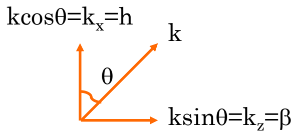
A multimode fiber
It has a larger radius and contains a large number of modes.

Fiber Optic Structure



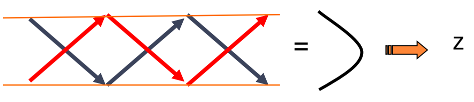
In standard optical fibers the core material is a highly pure silica glass (SiO_2) compound and is surrounded by a glass cladding

Ray Picture



- **k is the ray propagation constant**
- **k=h a_x+β a_z**
- $e^{j(k.r)} = e^{j(hx+\beta z)}$
- **2 propagating plane waves = standing wave pattern**

$$e^{-j(hx+\beta z)} + e^{j(hx-\beta z)} = 2\cos(h\,x)e^{-j\beta z}$$



Ray Picture

- **2 plane waves with an incidence angle θ = electric field distribution in the transverse plane which propagates in the z direction (longitudinal direction) with a propagation constant β.**
- **For a specific values of θ, this electric field distribution does not change during propagation and it is called mode.**
- **Each mode has a different electric field distribution and a propagation constant β.**
- **A single-mode fiber sustains only one mode of propagation, whereas multimode fibers contain a large number of modes.**

Linearly Polarized modes (LP modes)

***Some of the fiber modes have the same propagation constant β. This means that they propagate with the same velocity. We can then get the sum of these modes and obtain a new different mode with a new electric field distribution. This new mode called Linearly Polarized (LP) mode.**

***It is called Linearly Polarized mode as the direction of the electric field does not change with time.**

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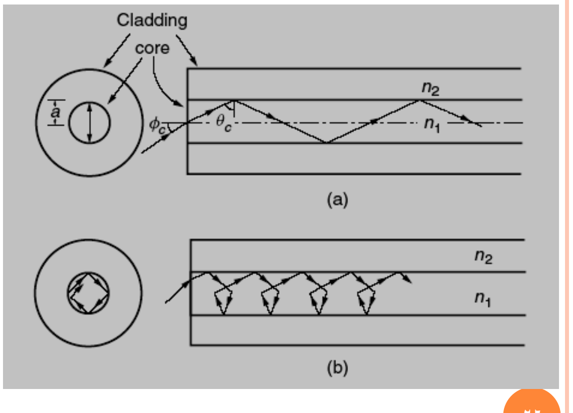
25

Ray types

***Meridional rays (pass by the fiber axis)**

***Skew rays**



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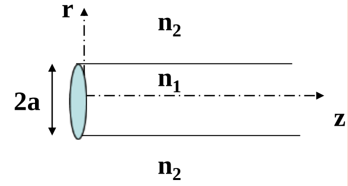
26

Structure of Step-Index Fibers

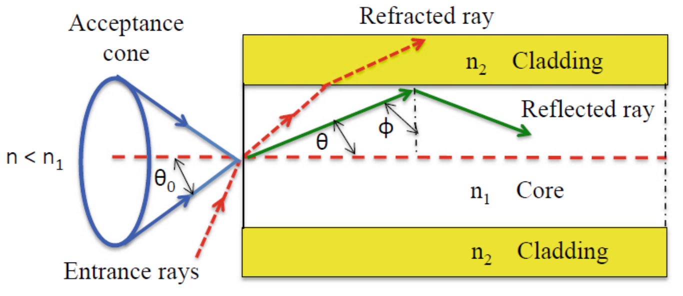
* In practical step-index glass fibers, the core of radius *a* has a refractive index n_1. This is surrounded by a cladding of slightly lower index n_2, where

$$n(r) \begin{cases} = n_1 & \text{for } |r| < a \\ = n_2 & \text{for } |r| > a \end{cases}$$



$$n_2 = n_1(1-\Delta)$$

$$\Delta = \frac{n_1 - n_2}{n_1}$$



Acceptance angle

Critical angle ϕ_c $\qquad$ $sin\ \phi_c = \dfrac{n_2}{n_1}$ $\qquad$ (1)

*Rays striking the core-to-cladding interface at angles less than φc will refract out of the core and be lost in the cladding, as the dashed line shows. The optical wave associated to these angles are called radiation modes

*Modes associated with incident angles greater than φc are called guided modes

*By applying Snell's law to the air–fiber face boundary, the condition of Eq. (1) can be related to the maximum entrance angle $\theta_{0,\ max}$, which is called the *acceptance angle* θ_A, through the relationship

$$n\ sin\ \theta_{0,max} = n\ sin\ \theta_A = n_1 sin\ \theta_c$$

$$\theta_c = \pi/2 - \varphi_c.$$

Numerical Aperture NA

$$n\sin\theta_A = n_1\cos\varphi_c = n_1\sqrt{1-\sin^2\varphi_c} = \sqrt{n_1^2 - n_2^2}$$

→ Acceptance angle θ_A which is the maximum angle with the normal to the air-core interface that corresponds to the critical incidence angle is related to the refractive indices by

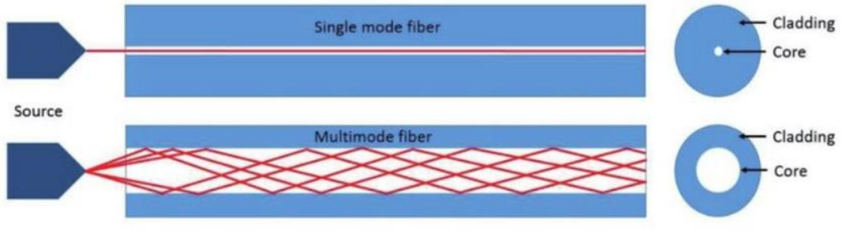
$$n\sin\theta_A = \sqrt{n_1^2 - n_2^2} = \mathrm{NA}$$

$$NA \approx n_1\sqrt{2\Delta}$$

The approximation on the right-hand side is valid for the typical case where , Δ is much less than 1.

For the rays to propagate in the core and does not refract in the cladding, they must be incident in the air-core interface within the acceptance cone.

Single mode and multimode fiber



Multimode → multiple angle of incidence

For the multimode fiber, the higher order modes are the modes with incidence angles close to the critical angle and lower order modes are the modes with incidence angles far from the critical angle

Advantages of multimode fiber

*The larger core radii of multimode fibers make it easier to launch optical power into the fiber and facilitate the connecting together of similar fibers.

*Another advantage is that light can be launched into a multimode fiber using a light-emitting-diode (LED) source, whereas single-mode fibers must generally be excited with laser diodes.

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31

Disadvantages of multimode fiber

For high-speed long-distance transmission, the bandwidth of multimode fiber is restricted by intermodal dispersion. Briefly, intermodal dispersion can be defined as follows:

When an optical pulse is launched into a fiber, the optical power in the pulse is distributed over all (or most) of the modes of the fiber. Each of the modes that can propagate in a multimode fiber travels at a slightly different velocity. This means that the modes in a given optical pulse arrive at the fiber end at slightly different times, thus causing the pulse to spread out in time as it travels along the fiber. This effect is known as *intermodal dispersion* or *modal delay* and can be reduced by using a graded-index profile in a fiber core.

Higher bandwidths are possible in single-mode fibers, where intermodal dispersion effects are not present because only one mode travels in the fiber.

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32

V number and number of modes in a fiber

An important dimensionless parameter that is related to the wavelength, core radius, and the numerical aperture is defined. It is called the *normalized frequency V* (also called the V *number* or V *parameter*) and defined by

$$V = \frac{2\pi a}{\lambda}(n_1^2 - n_2^2)^{1/2} \approx \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

This V number determines how many modes a fiber can support through the following relation (if the number of modes is large)

$$M = \frac{1}{2}\left(\frac{2\pi a}{\lambda}\right)^2 (n_1^2 - n_2^2) = \frac{V^2}{2}$$

where M is the number of modes in the fiber.

V number and number of modes in a fiber

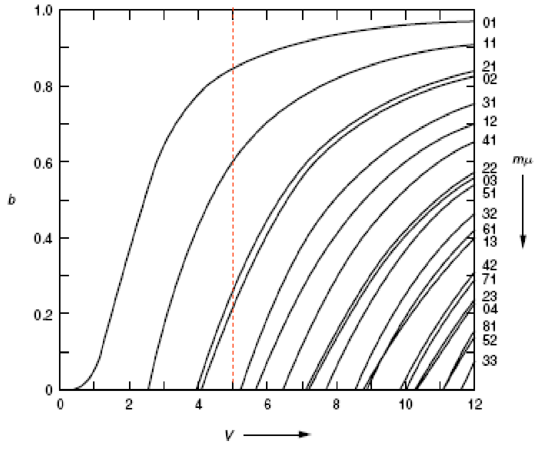
*For a few number of modes, another method is used to calculate the number of modes.

We define another dimensionless parameter called ***normalized propagation constant b***

$$b = \frac{(\beta/k)^2 - n_2^2}{n_1^2 - n_2^2}$$

Then a plot between b and V is used to determine the number of modes

V number and number of modes in a fiber



Each line in this curve represents the V-b relation for a different LP mode.

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35

V number and number of modes in a fiber

*This V-b figure shows that except for the lowest-order mode LP01, each mode can exist only for values of V that exceed a certain limiting value (with each mode having a different V limit).

*To calculate the number of modes, for a few mode fiber, we calculate the V number and then draw a vertical line at this value and calculate the number of intersection points (each intersection point represents a mode).

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36

V number and cutoff wavelength

The principle on which single-mode fibers are based is by appropriately choosing core radius a, n_1, and n_2 so that

$$V = \frac{2\pi a}{\lambda}\left(n_1^2 - n_2^2\right)^{1/2} \leq 2.405$$

Below this value of V number , all modes are cutoff except the fundamental mode. The fundamental mode LP01 (HE_{11}) has no cutoff and ceases to exist only when the core diameter is zero.

The minimum wavelength at which all modes are cut off except the fundamental mode is called the ***cutoff wavelength*** λ_c.

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37

V number and cutoff wavelength

Using the definition of V number

$$V = \frac{2\pi a}{\lambda}(n_1^2 - n_2^2)^{1/2} \approx \frac{2\pi a}{\lambda} n_1\sqrt{2\Delta}$$

And with the limiting value of V number for the single mode operation of 2.405, the cutoff wavelength could be deduced as

$$\lambda_c = \frac{2\pi a}{2.405}(n_1^2 - n_2^2)^{1/2}$$

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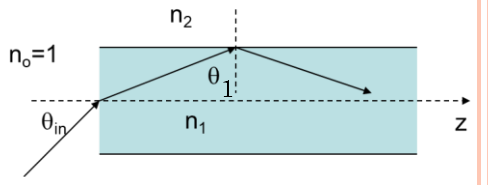
38

Effective refractive index n_{eff}

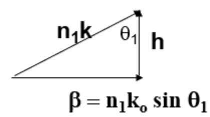
For some fiber mode, the propagation constant β is a factor of n_{eff} times the vacuum wave number $k_0 = 2\pi/\lambda$, that is,

$$n_{eff} = \frac{\beta}{k_0}$$



$$\beta = n_{eff} k_o$$

$$n_{eff} = n_1 \sin\theta_1$$



$\pi/2 > \theta_1 > \theta_c$

$\rightarrow$ $n_1 > n_{eff} > n_2$

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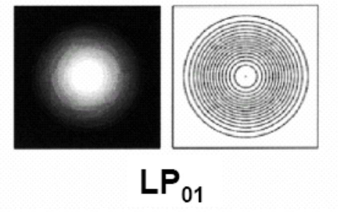
Fundamental mode $HE_{11}=LP_{01}$

Useful Approximation (Gaussian distribution)

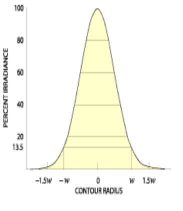
$E(r)=E_o \exp[-(r/w)^2]$

where $w=a\,[0.65+1.619V^{-3/2}+2.87V^{-6}]$



Mode field diameter:

The diameter at which the electric field of the fundamental mode drops to 1/e from its maximum.



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